

Arjuna

JEE 2024



- Subject Name- Physics

- Chapter Name- Simple Harmonic Motion

-01



Lecture No.-

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Today's Goal

1

SHM Equation of SHM

2

3

4

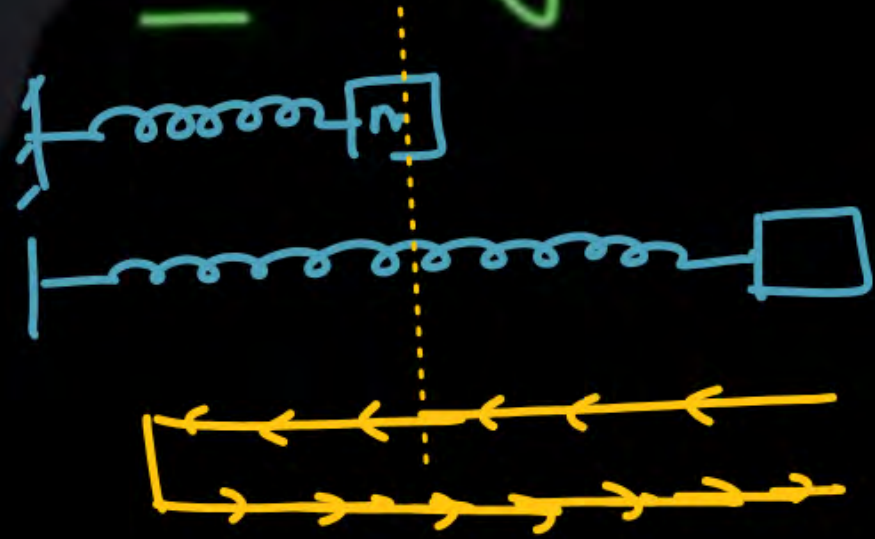


Periodic motion $\rightarrow$ Any motion which repeats itself after regular interval of time.

The const interval of time after which the motion is repeated is called time period.

Ex $\rightarrow$ spring balance system, motion of earth around sun,
Uniform circular motion, pendulum

Oscillatory motion



- motion is oscillatory if it moves to & fro about a fixed point after regular interval of time

SHM → It is a special kind of oscillatory motion in which net force on particle at any time (any vto) is proportional to negative of displacement from mean position
 $\rightarrow \vec{F}_{\text{net}} = 0$
 or Equil. position

$$\vec{F}_{\text{net}} \propto -\vec{x}$$

$$\vec{F}_{\text{net}} = -k \vec{x}$$

net ext force

SHM का const

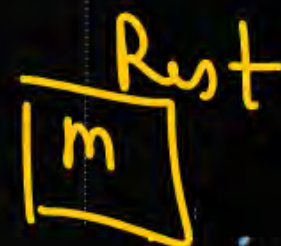
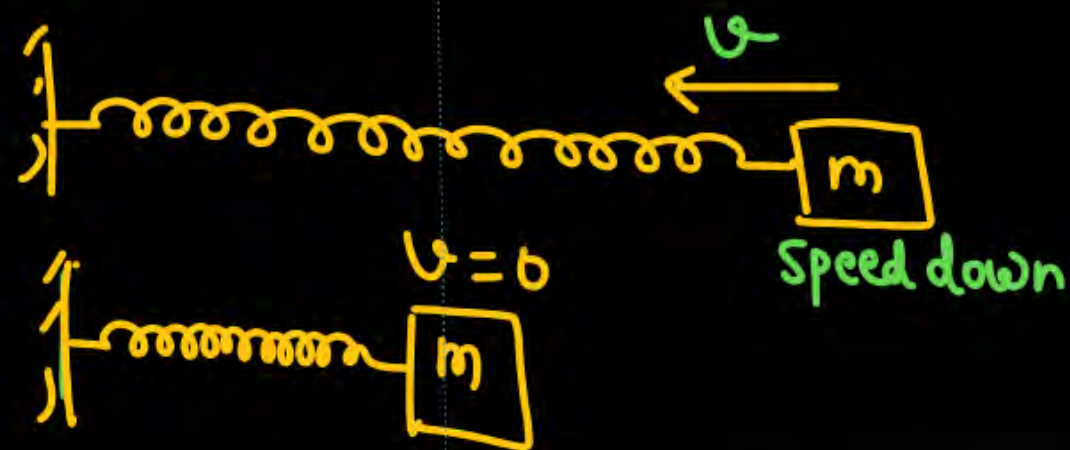
displ. of particle from mean position (m.p.)

At m.p. $\Rightarrow \vec{F}_{\text{net}} = 0$

$$\vec{F}_{\text{net}} \propto -\vec{x}$$

$$\vec{a}_{\text{net}} \propto -\vec{x}$$

$$\vec{a}_{\text{net}} = -\frac{k}{m} \vec{x}$$

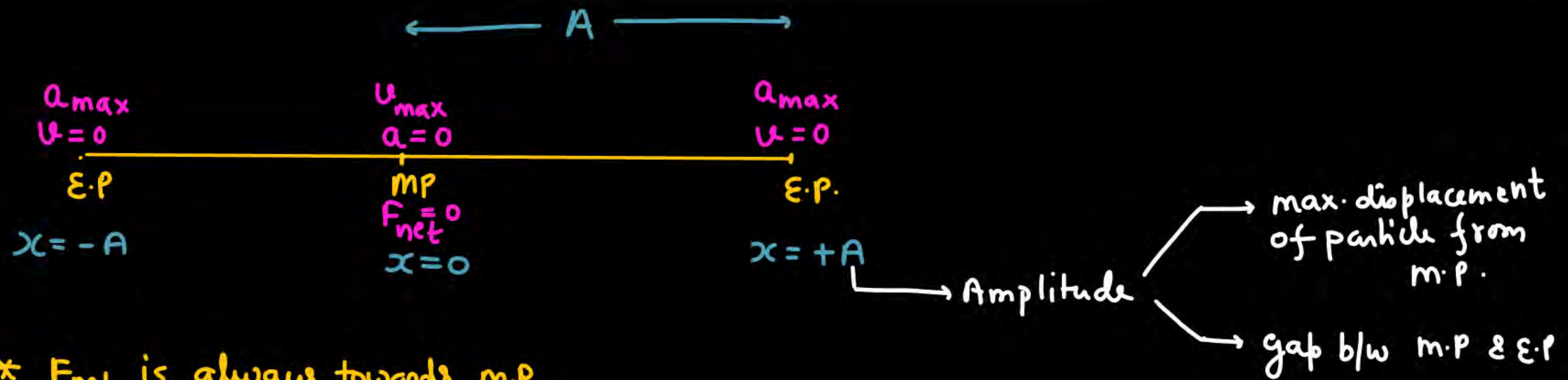


$$\vec{F}_{net} = \cancel{\vec{F}_{mg}} + \cancel{\vec{F}_N} + \vec{F}_{sp}$$

$$F_{net} = kx \text{ (magnitude)}$$

$$\boxed{\vec{F}_{net} = -k\vec{x}} \rightarrow \text{SHM} \checkmark$$

$$\vec{a} = -\frac{k}{m}\vec{x}$$



* F_{net} is always towards M.P.

* Acc " " " "

* particle will speed up when move towards M.P. (towards)

& will speed down " " away from M.P. (from)

Eqⁿ of SHM

$$\vec{F}_{\text{net}} = -k\vec{x}$$

$$\vec{a} = -\frac{k}{m}\vec{x}$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

Solve & get

$$x = A \sin(\omega t + \phi)$$

displacement
↑
 $x = A \sin(\omega t + \phi)$
Amplitude

$$v = \frac{dx}{dt} = A\omega \cos(\omega t + \phi)$$

$$a = \frac{dv}{dt} = -A\omega \sin(\omega t + \phi) \times \omega$$

$$a = -\omega^2 \underline{A \sin(\omega t + \phi)}$$

$$a = -\omega^2 x$$

$$\vec{a} = -\underline{\omega^2} \vec{x}$$

$$\vec{a} \propto -\vec{x}$$

$$\vec{F}_{\text{net}} \propto -\vec{x}$$

$$\vec{F}_{\text{net}} = -k\vec{x}$$

$$\frac{k}{m} = \omega^2 \quad k = m\omega^2$$

$$\omega = \sqrt{\frac{k}{m}} = \frac{2\pi}{T}$$

Time period

$$T = 2\pi \sqrt{\frac{m}{k}}$$

Time period of SHM

$$\vec{F} = -k\vec{x} \xrightarrow{\text{sol}^n} \vec{x} = A \sin(\omega t + \phi)$$

Displacement of particle from m.p.

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$k = m\omega^2$$

$$\omega = \frac{2\pi}{T}$$

Angular freq.
Time period.

$A \rightarrow$ Amplitude

$\omega \rightarrow$ Angular freq. $= \omega = \frac{2\pi}{T}$

$(\omega t + \phi) \rightarrow$ phase

$\phi \rightarrow$ initial phase

Q A particle is performing SHM s.t its displacement from m.p. at any time is given as

$$x = 10 \sin(4\pi t + \pi/6)$$

Find

① Amplitude = 10

$$x_{\max} = 10$$

② $\omega = ?$ $4\pi = \omega$

③ Time period $\omega = \frac{2\pi}{T}$

$$4\pi = \frac{2\pi}{T}$$

$$T = \frac{1}{2} \text{ sec}$$

④ phase = $4\pi t + \pi/6$

⑤ phase at $t = \frac{1}{4} \text{ sec}$

$$\begin{aligned} \text{phase} &= 4\pi \times \frac{1}{4} + \frac{\pi}{6} \\ &= 7\pi/6 \end{aligned}$$

⑥ phase at $t=0$ phase = $\pi/6$
Initial phase

⑦ find displacement of particle from m.p. at $t=0$

$$x = ?$$

$$x = 10 \sin(0 + \pi/6) = 5$$

⑧ $x = 10 \sin(4\pi t + \pi/6)$

find displacement of particle at $t = \frac{1}{4}$ sec

$$x = 10 \sin\left(4\pi \times \frac{1}{4} + \frac{\pi}{6}\right)$$

$$= 10 \sin(180 + 30)$$

$$x = -5$$

⑨ find velocity of particle at $t = \frac{1}{4}$ sec

$$v = \frac{dx}{dt} = 10 \times 4\pi \cos(4\pi t + \pi/6)$$

$$= 40\pi \cos(4\pi t + \pi/6)$$

Q A particle is performing SHM s.t its displacement from m.p. is given as

$$x = 10 \sin\left(\frac{\pi t}{3} + \frac{\pi}{6}\right) \text{ Find}$$

① $A = 10$

② $\omega = \frac{\pi}{3} = \frac{2\pi}{T}$

③ $T = 6$

④ initial phase = $\pi/6$

⑤ Displacement of particle at $t = 0$

$$x = 10 \sin(0 + \pi/6) = 5$$

⑥ Displacement of particle at $t = 1 \text{ sec}$

$$x = 10 \sin\left(\frac{\pi}{3} \times 1 + \frac{\pi}{6}\right) \Rightarrow x = 10 \quad x = +A$$

⑦ $v = f(t)$

$$v = \frac{10\pi}{3} \cos\left(\frac{\pi t}{3} + \frac{\pi}{6}\right)$$

⑧ $a = f(t)$

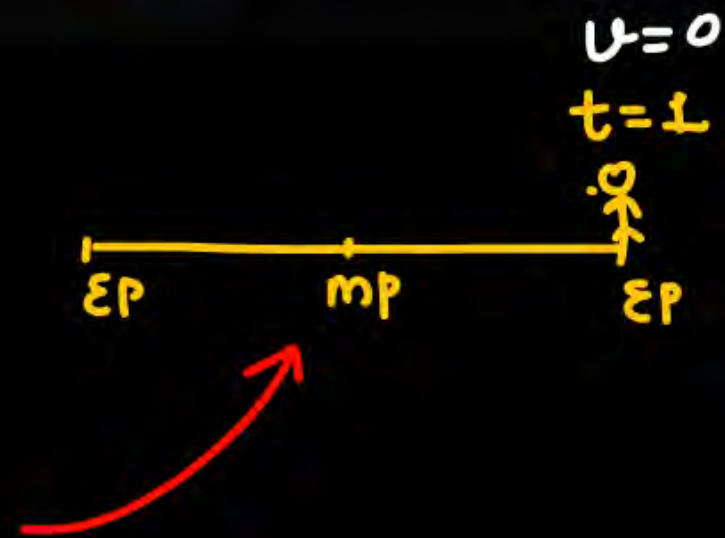
$$a = -\frac{10\pi^2}{9} \sin\left(\frac{\pi t}{3} + \frac{\pi}{6}\right)$$

⑨ v at $t = 1 \text{ sec}$

$$v = 0$$

⑩ a at $t = 1 \text{ sec}$

$$a = -\frac{10\pi^2}{9}$$



8:30 pm

Q A particle is performing SHM having amplitude A , angular freq. ω s.t. at $t=0$ particle is at $x = +A/2$ moving away from m.p. Find eqⁿ of SHM.

Solⁿ

$$x = A \sin(\omega t + \phi)$$

$$\phi = ?$$

put $t=0, x = A/2$

$$\frac{A}{2} = A \sin(0 + \phi)$$

$$\sin \phi = \frac{1}{2}$$

$$\phi = 30^\circ, 150^\circ$$

at $t=0, v > 0$

$$v = \frac{dx}{dt} = A\omega \cos(\omega t + \phi) > 0$$

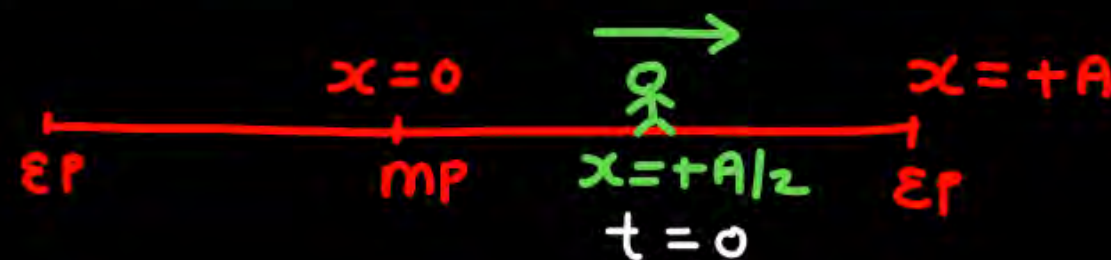
$$t=0, v = A\omega \cos(0 + \phi) > 0$$

$$\cos \phi > 0$$

$$\phi = 30^\circ$$

$$\phi = 150^\circ \times$$

$$x = A \sin(\omega t + 30^\circ)$$



8:30 PM

Q A particle is performing SHM having amplitude A , angular freq. ω s.t. at $t=0$ particle is at $x = +A/2$ moving towards from m.p. Find eqⁿ of SHM.

Solⁿ $x = A \sin(\omega t + \phi)$

$\phi = ?$

put $t=0, x = A/2$

$\frac{A}{2} = A \sin(0 + \phi)$

$\sin \phi = \frac{1}{2}$

$\phi = 30^\circ, 150^\circ$

at $t=0, v < 0$

$v = \frac{dx}{dt} = A\omega \cos(\omega t + \phi) < 0$

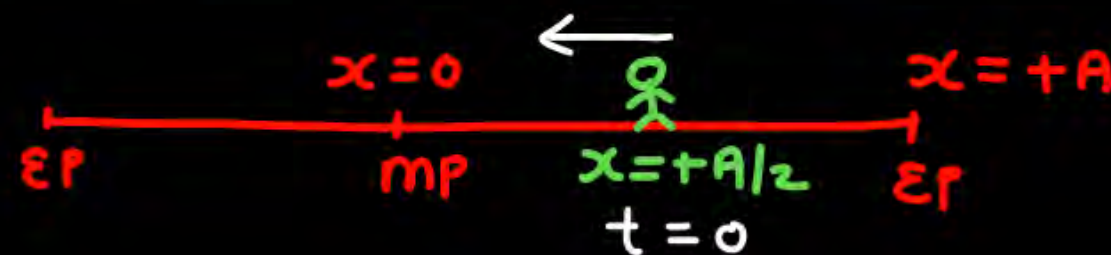
$t=0, v = A\omega \cos(0 + \phi) < 0$

$\cos \phi < 0$

$\phi = 30^\circ$ X

$\phi = 150^\circ$ ✓

$x = A \sin(\omega t + 150^\circ)$



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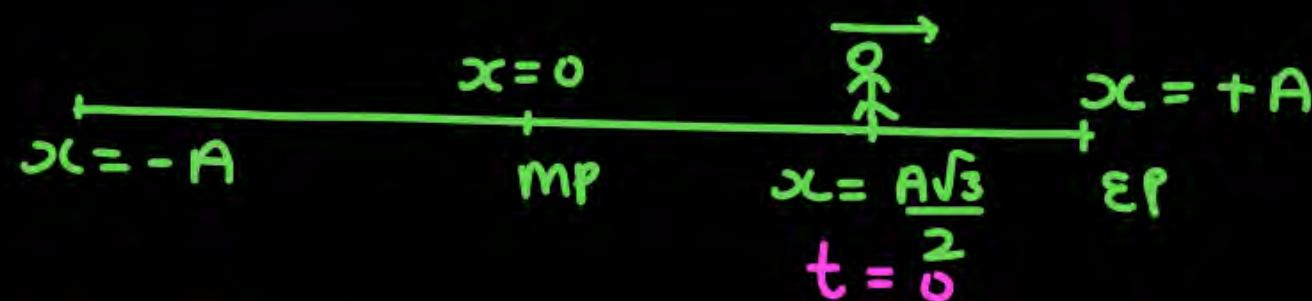
Q A particle is performing SHM having amplitude A , angular freq. ω s.t. at $t=0$ particle is at $x = \frac{A\sqrt{3}}{2}$ moving away from m.p. Find eqⁿ of SHM.

Solⁿ $x = A \sin(\omega t + \phi)$

$t=0, x = \frac{A\sqrt{3}}{2}$

$\frac{A\sqrt{3}}{2} = A \sin(0 + \phi)$

$\phi = 60^\circ, 120^\circ$



$t=0, v > 0$

$v = A\omega \cos(\omega t + \phi) > 0$ at $t=0$

$v = A\omega \cos(0 + \phi) > 0$

$\cos \phi > 0$

$\phi = 60^\circ$

$\phi = 120^\circ \times$

$x = A \sin(\omega t + 60^\circ)$

8:30 pm

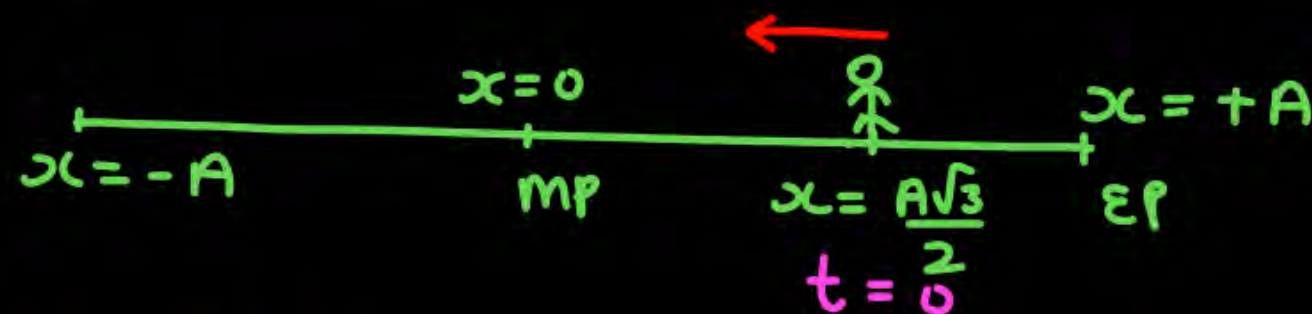
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Solⁿ $x = A \sin(\omega t + \phi)$

$t=0, x = \frac{A\sqrt{3}}{2}$

$\frac{A\sqrt{3}}{2} = A \sin(0 + \phi)$

$\phi = 60^\circ, 120^\circ$



$t=0, v < 0$

$v = A\omega \cos(\omega t + \phi) < 0$ at $t=0$

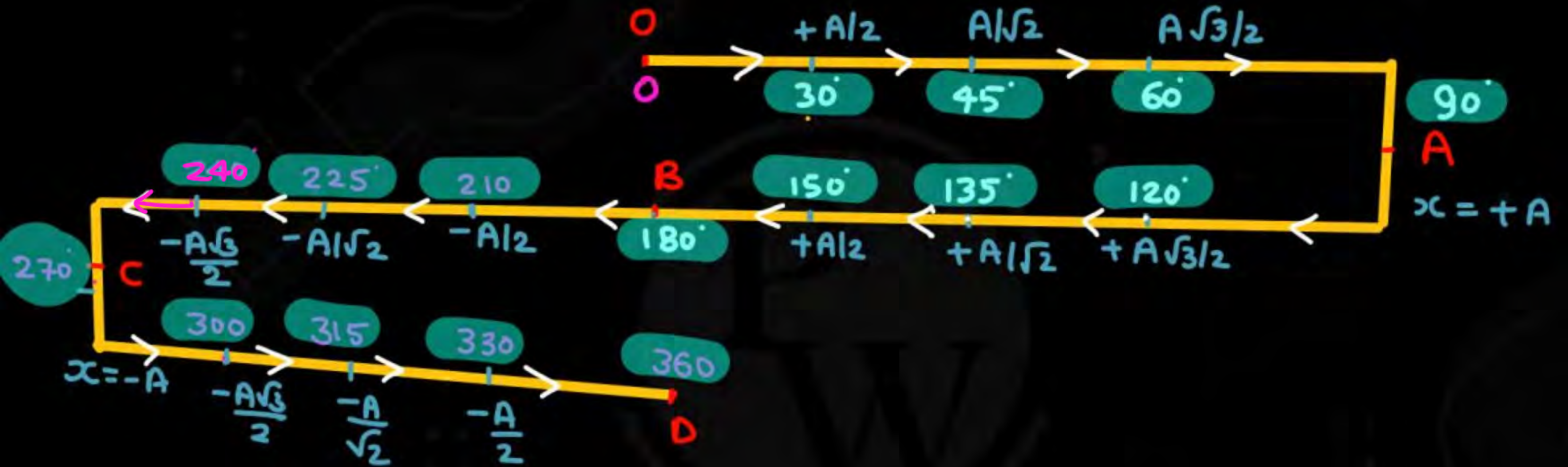
$v = A\omega \cos(0 + \phi) < 0$

$\cos \phi < 0$

$\phi = 60^\circ$ X

$\phi = 120^\circ$ ✓

$x = A \sin(\omega t + 120^\circ)$



Home Work

- Complete your Backlog's
- 3 PYQ Attached + today's ques solve

Question



A particle executes simple harmonic motion represented by displacement function as $x(t) = A \sin(\omega t + \phi)$

If the position and velocity of the particle at $t = 0$ s are 2 cm and 2ω cm s⁻¹ respectively, then its amplitude is $x\sqrt{2}$ cm where the value of x is_____.

[JEE Mains 2021]

Ans. (2)

Question



The equation of a particle executing simple harmonic motion is given by $x = \sin\pi\left(t + \frac{1}{3}\right)$ m. At $t = 1$ s, the speed of particle will be:
(Given : $\pi = 3.14$)

[JEE Mains 2021]

Ans. (157)

Question



$Y = A \sin(\omega t + \phi_0)$ is the time-displacement equation of a SHM. At $t = 0$ the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x-direction. Then the initial phase angle ϕ_0 will be:

[JEE Mains 2021]

- A** $\frac{\pi}{6}$
- B** $\frac{\pi}{3}$
- C** $\frac{5\pi}{6}$
- D** $\frac{2\pi}{3}$

Ans. (C)

thank you

